Project Executable Files

Date	7 July 2026
Team ID	
Project Name	Smart Agricultural Production Optimization Engine
Maximum Marks	3 Marks

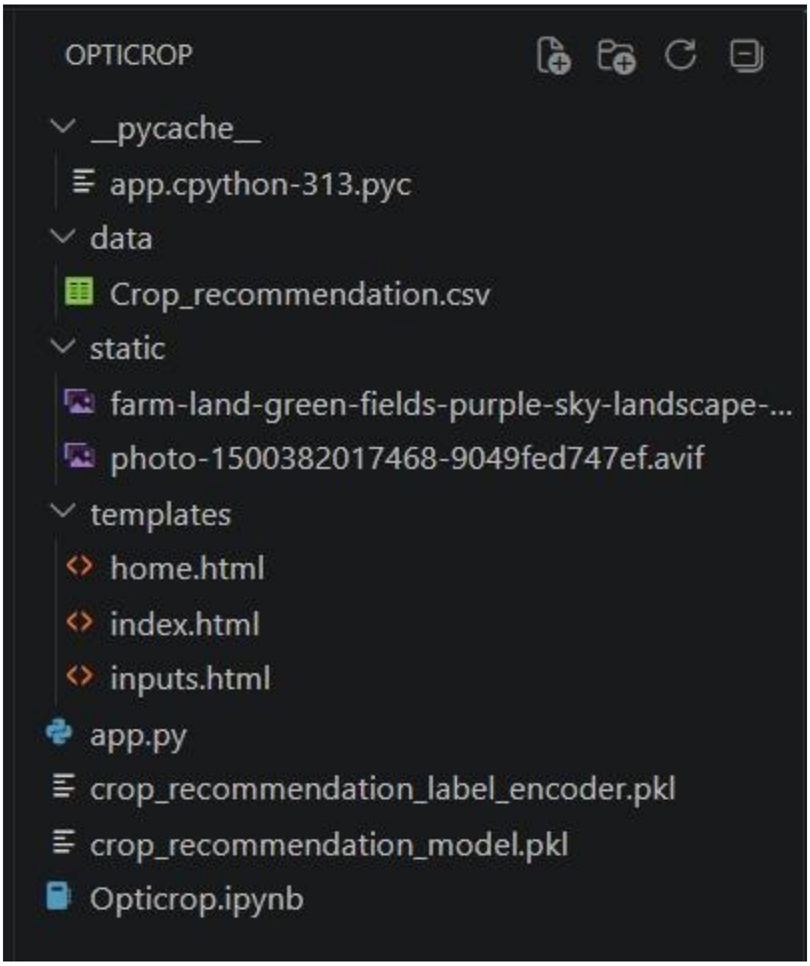
Project Executable Files

This document contains all the executable and deployable components of the **Smart Agricultural Production** project. The project is developed using Python, Flask, HTML, CSS, JavaScript, and Machine Learning. All source files, datasets, trained model files, templates, static resources, and dependency files are organized in a structured directory, allowing evaluators to run the application with minimal setup.

Step 1: Submission Checklist

S.No	Item to Submit	Submitted (Yes / No / NA)
1	Complete source code (all files and folders)	**Yes**
2	README / Setup Guide (instructions to run the project)	**Yes**
3	requirements.txt / dependency file	**Yes**
4	Database schema / seed files (if applicable)	**NA**
5	Environment configuration file (.env.example or similar)	**NA**
6	Deployed application URL (if hosted)	**No**
7	APK / Executable binary (if applicable)	**NA**
8	Dockerfile / Containerization configuration	**NA**
9	Test files and test results	**Yes**
10	Demo video or walkthrough	**Yes**

Step 2: File / Folder Structure



Step 3: Deployment / Access Details

Field	Details
Hosted / Deployed URL	Not Hosted (Runs Locally)
Login Credentials (Demo)	Not Required
Platform / Hosting Provider	Local Flask Development Server
Repository Link	Git-hub
Demo Video Link	Demo-link

Step 4: Run Instructions

Pre-requisites

- Python 3.10 or later
- Flask
- NumPy
- Pandas
- Scikit-learn
- Pickle (Python Standard Library)

Steps

Step 1: Download or clone the project source code.

Step 2: Open the project folder in Visual Studio Code or any Python IDE.

Step 3: Install the required dependencies using:

```
pip install -r requirements.txt
```

Step 4: Ensure that the trained machine learning model (`model.pkl`) and dataset are present in the project directory.

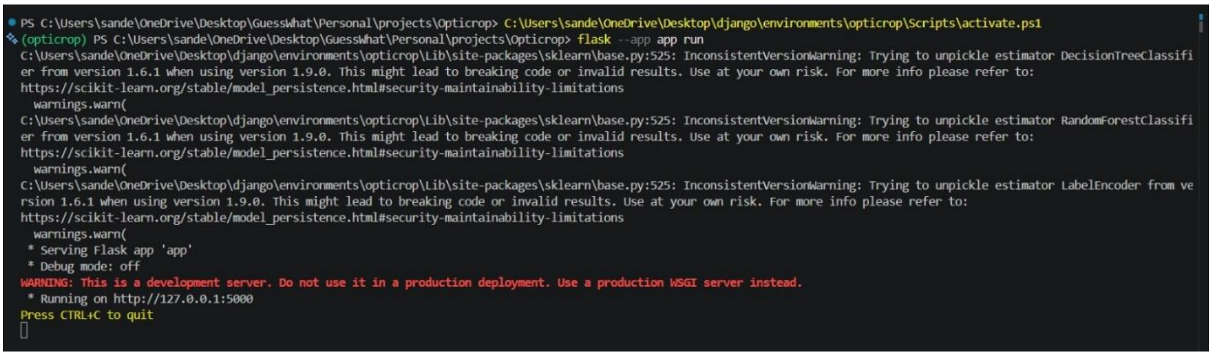
Step 5: Run the Flask application:

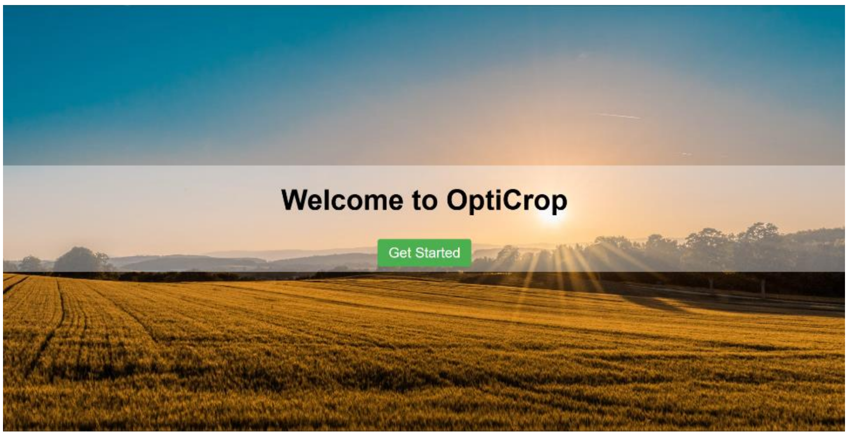
```
python app.py
```

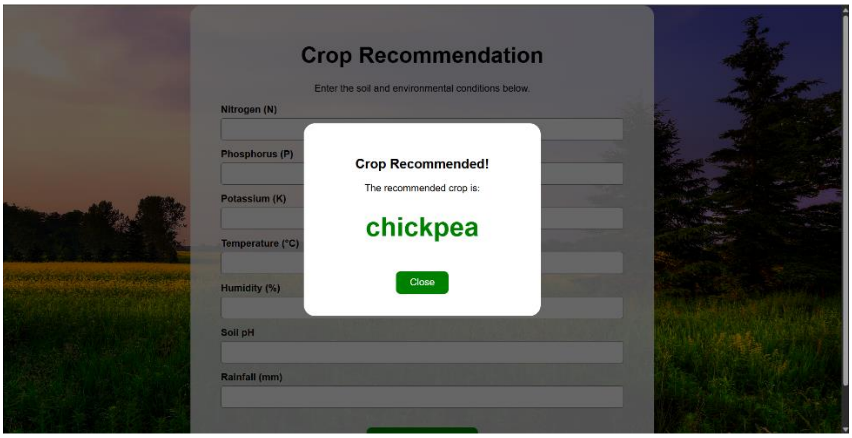
Step 6: Open a web browser and navigate to:

```
http://127.0.0.1:5000/
```

Step 7: Enter the required soil and environmental parameters, then click **Predict** to view the recommended crop.







Step 5: Known Issues / Limitations

S.No	Known Issue / Limitation	Workaround / Status
1	The application requires manually entered input values.	Future enhancement will integrate IoT sensors for automatic data collection.
2	Real-time weather information is not integrated.	Future enhancement will include Weather API integration.
3	The application currently runs on a local server only.	Can be deployed on cloud platforms such as Render, Railway, or PythonAnywhere in future versions.